Pablo Fernandez Carmona :: Center for proton therapy :: Paul Scherrer Institut

Continuous beam scanning intensity control of a medical proton accelerator using a Simulink generated FPGA gain scheduled controller

PCaPAC, Hsinchu, 19.10.2018

Outline

1. Introduction
2. Beam delivery challenges
3. Facility modelling
4. Controller design
5. Results
6. Conclusion

Introduction

Proton Therapy at the Paul Scherrer Institut

Cancer patient treatment areas:

- Gantry 1 (1996)
- Optis 2 (2010)
- Gantry 2 (2013)
- Gantry 3 (2018)

Gantry 2

Proton cyclotron

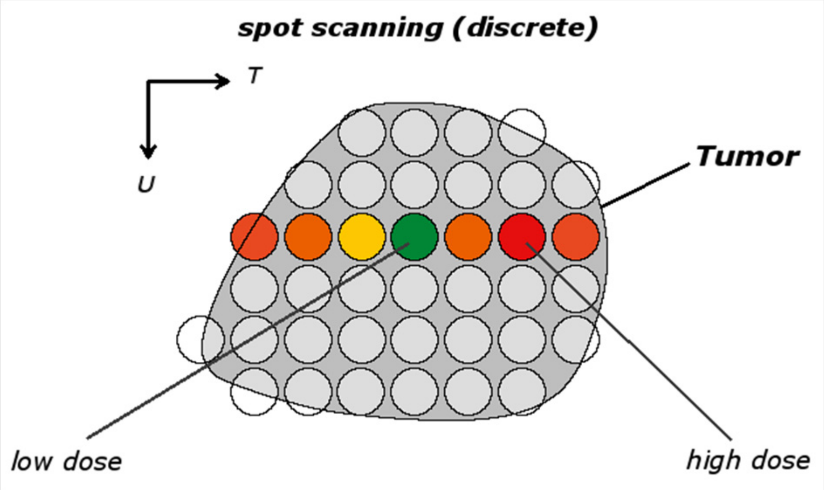
Introduction: beam delivery

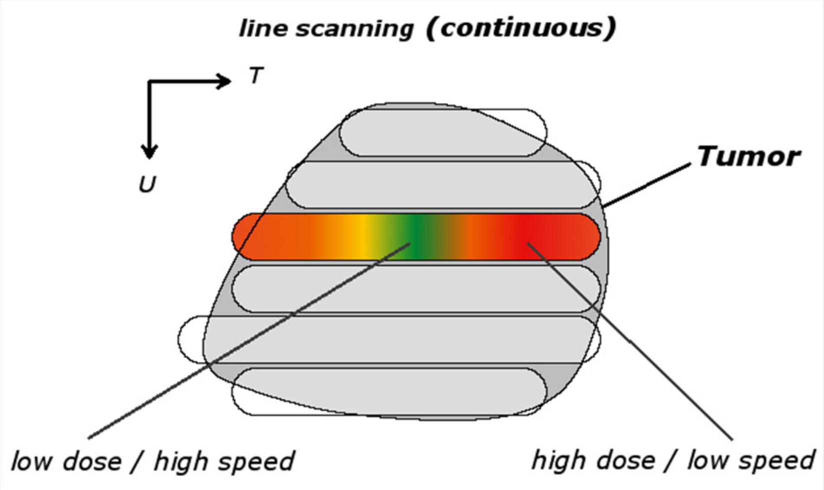
Spot scanning

- **Step and shoot**
- **Dose driven**: beam on until prescribed dose is reached
- Move to next position with beam off
- Beam instabilities do not affect delivery

Line scanning

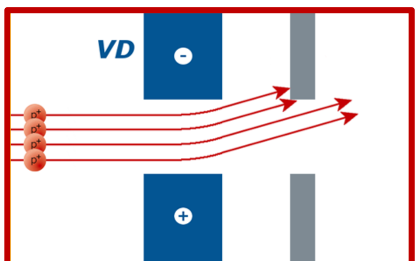
- **Continuous scanning**
- **Time driven**: moving while modulating beam
- Instantaneous over- or undercurrent cannot be compensated
- Requires precise beam current control
  - Reach target value quickly
  - No significant overshoot





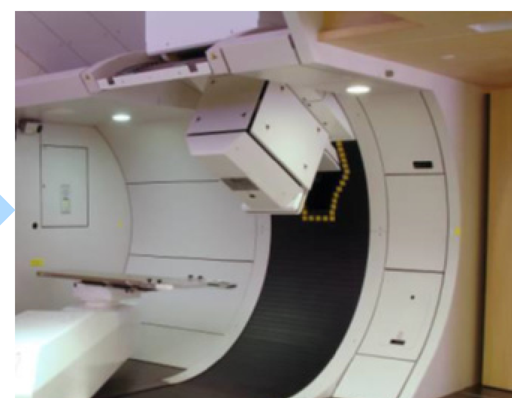
Courtesy V. Minnig

Introduction: Beam intensity control



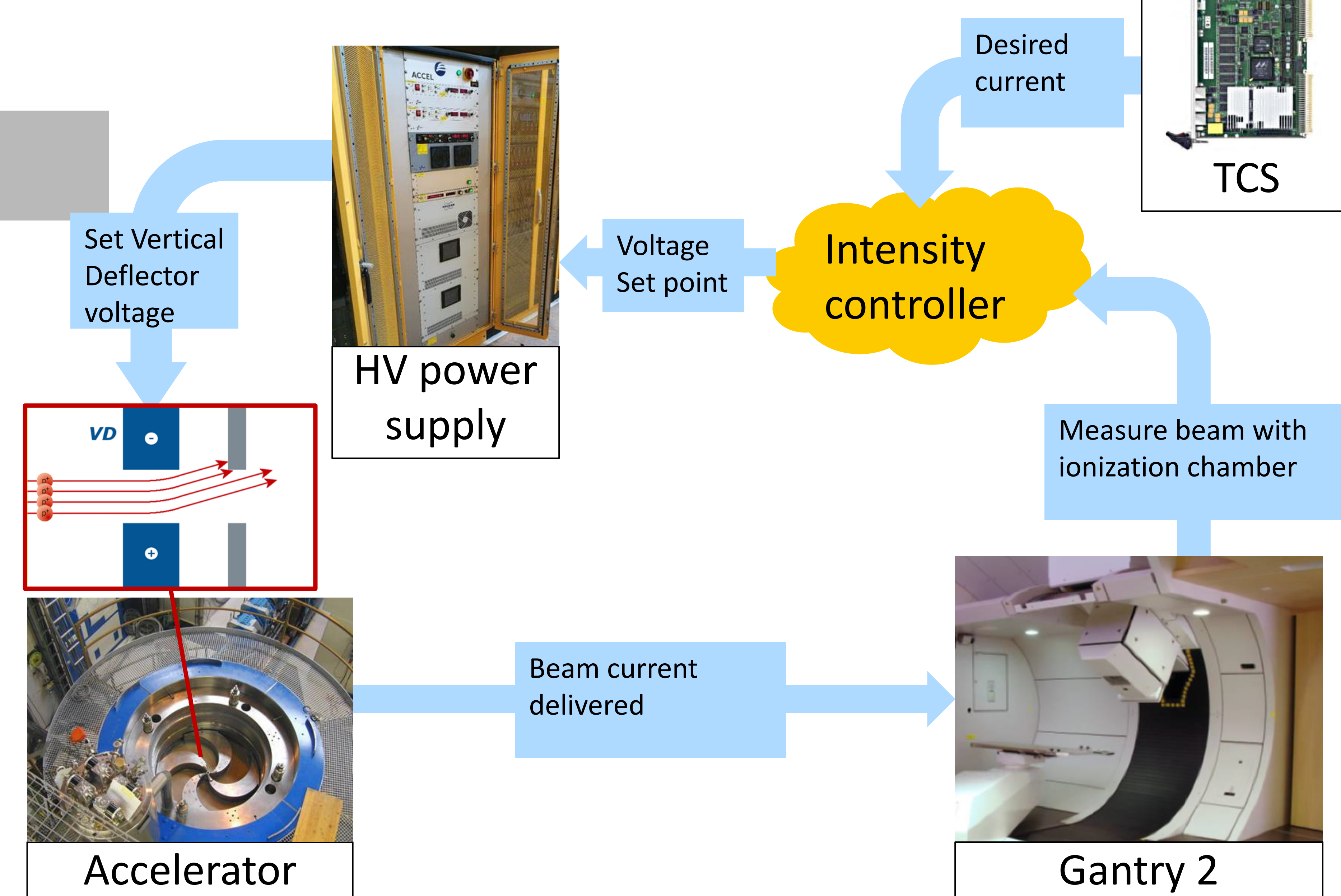
Accelerator

Beam current delivered



Gantry 2

Introduction: Beam intensity control



Delivery challenges I: The HW

- Open loop doesn't meet clinical specifications

Reach set point	50 µs
Remain stable (within 10V) from set point	100 µs
Voltage ripple	+-5 V
Maximum overshoot	10 V

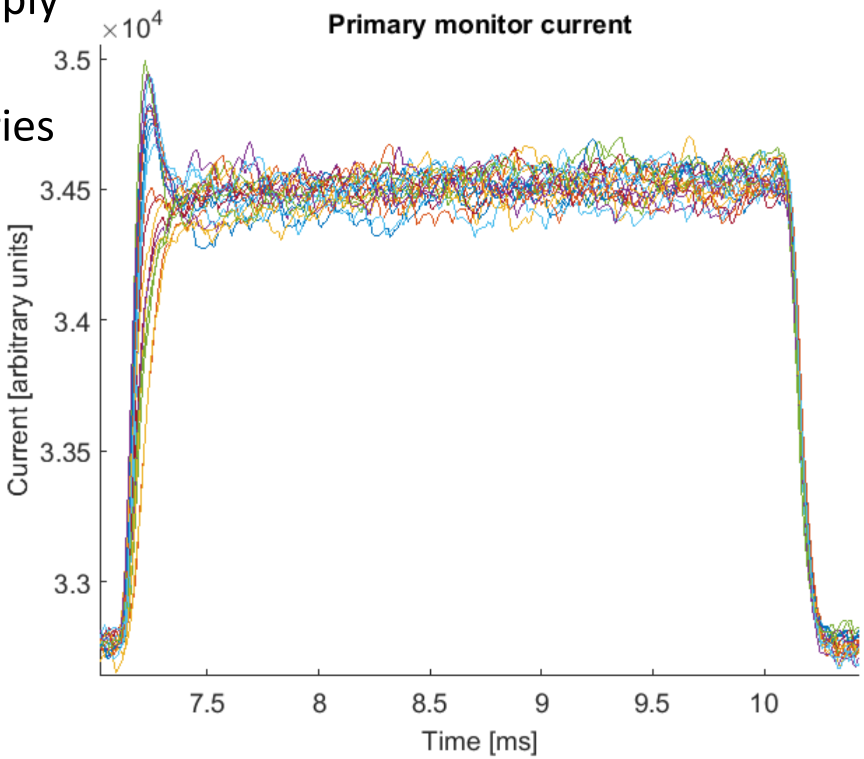
- Internal structure of the 5 KV power supply (not monolithic)
  - Spikes while crossing internal boundaries
  - Unpredictable step response of HV
  - Random beam overshoot after a jump

Delivery challenges I: The HW

- Open loop doesn't meet clinical specifications

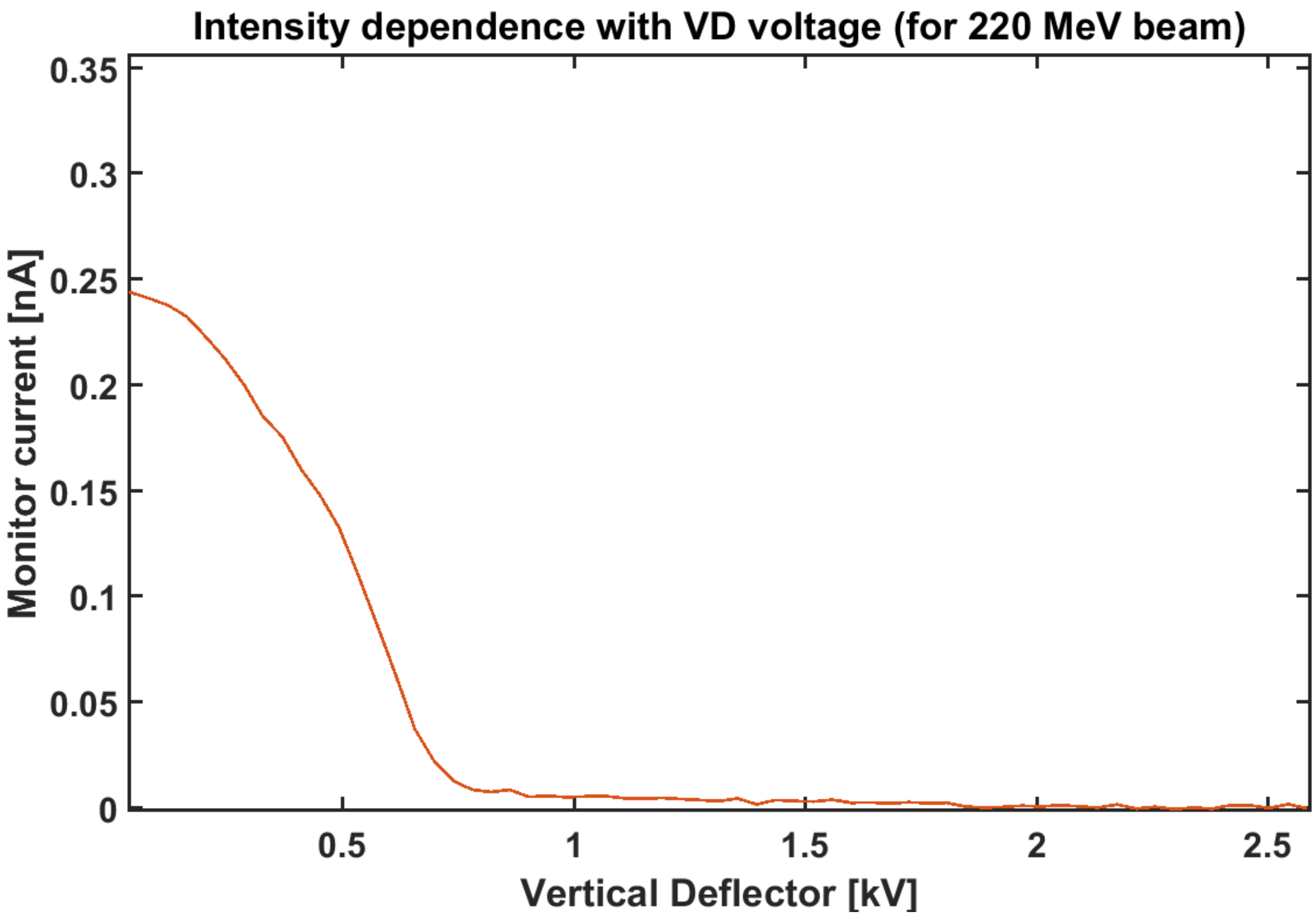
Reach set point	50 µs
Remain stable (within 10V) from set point	100 µs
Voltage ripple	+-5 V
Maximum overshoot	10 V

- Internal structure of the 5 KV power supply (not monolithic)
  - Spikes while crossing internal boundaries
  - Unpredictable step response of HV
  - Random beam overshoot after a jump



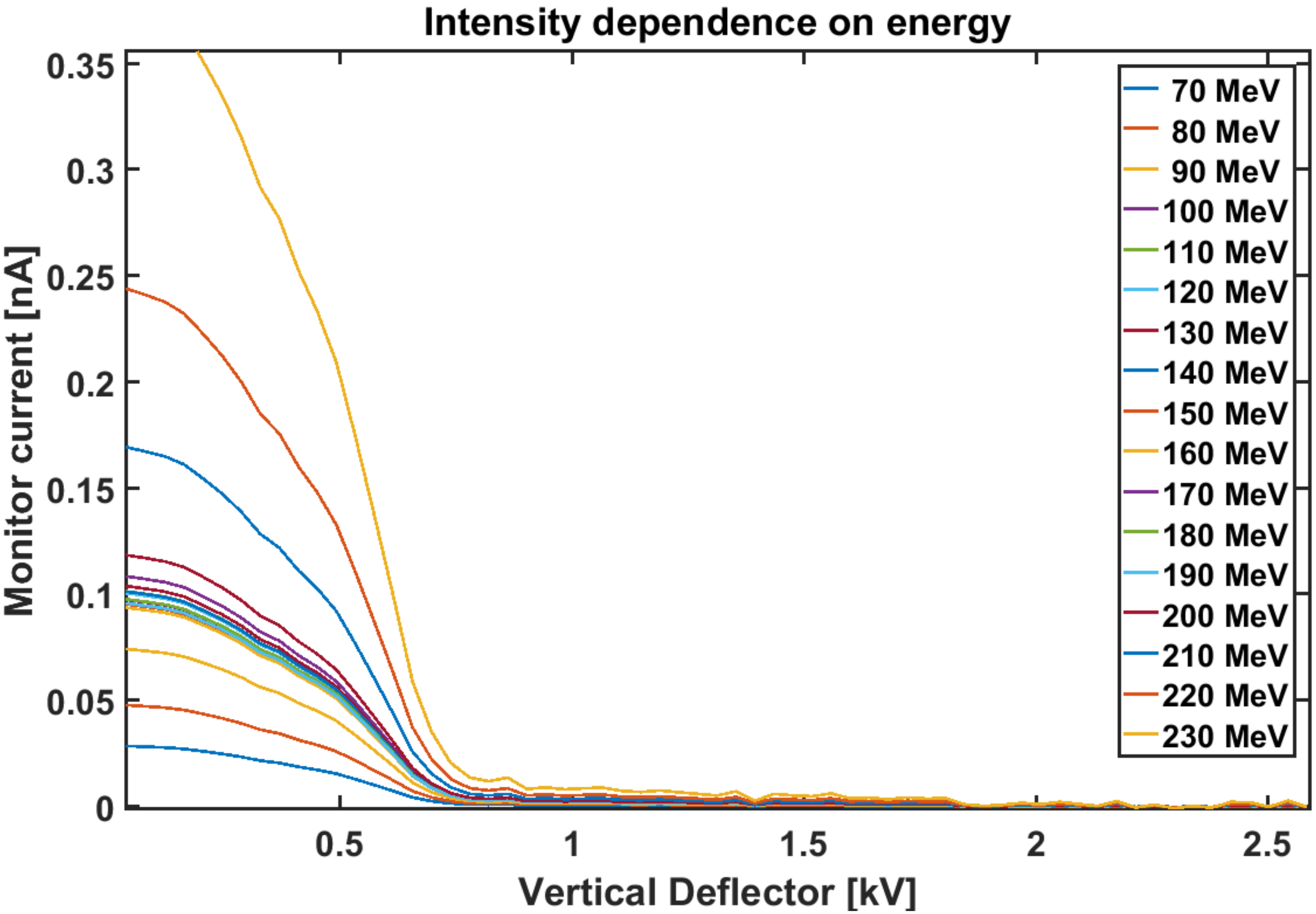
Delivery challenges: Accelerator

- Beam intensity non linear with VD voltage



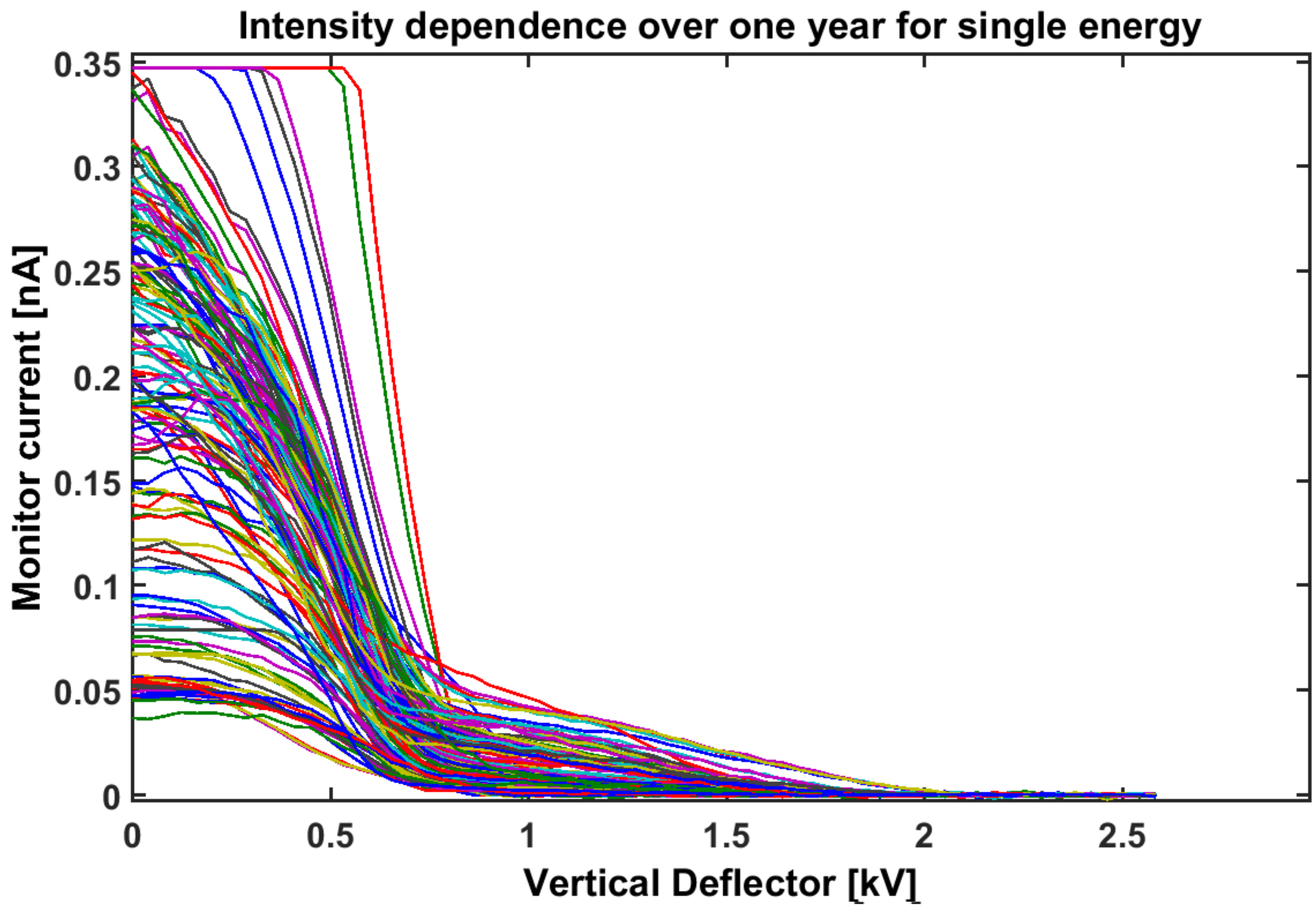
Delivery challenges: Accelerator

- Beam intensity non linear with VD voltage
- Energy dependence of the transfer function



Delivery challenges: Accelerator

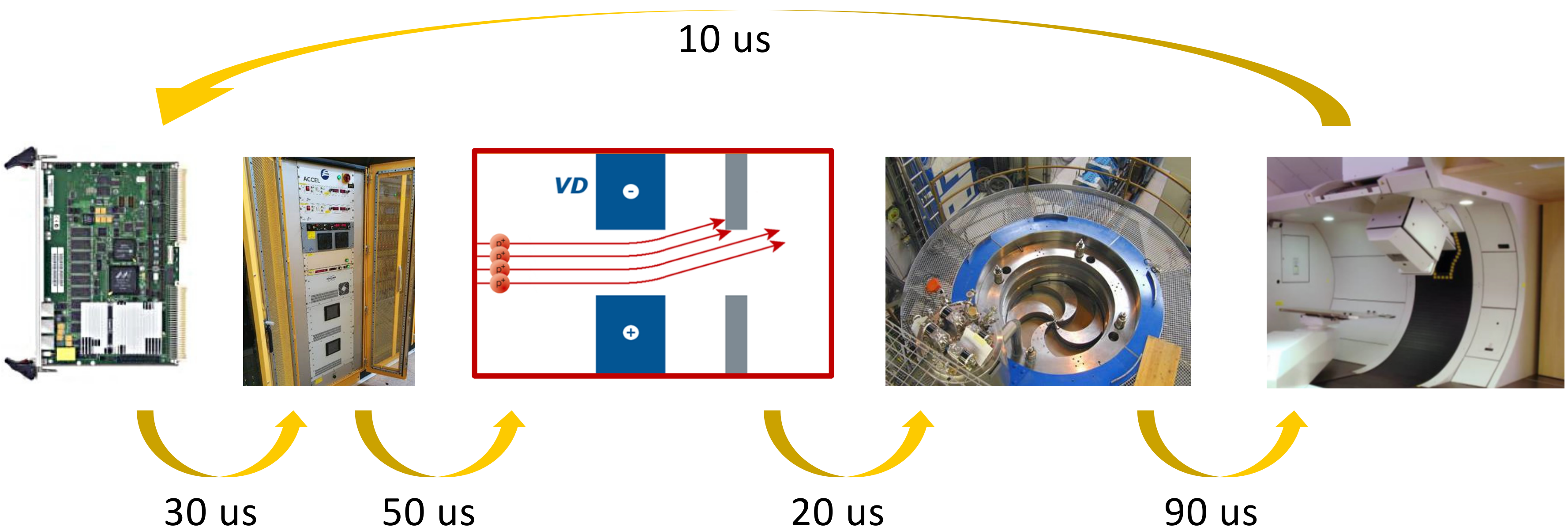
- Beam intensity non linear with VD voltage
- Energy dependence of the transfer function
- Variability of the transmissibility with time



PAUL SCHERRER INSTITUT

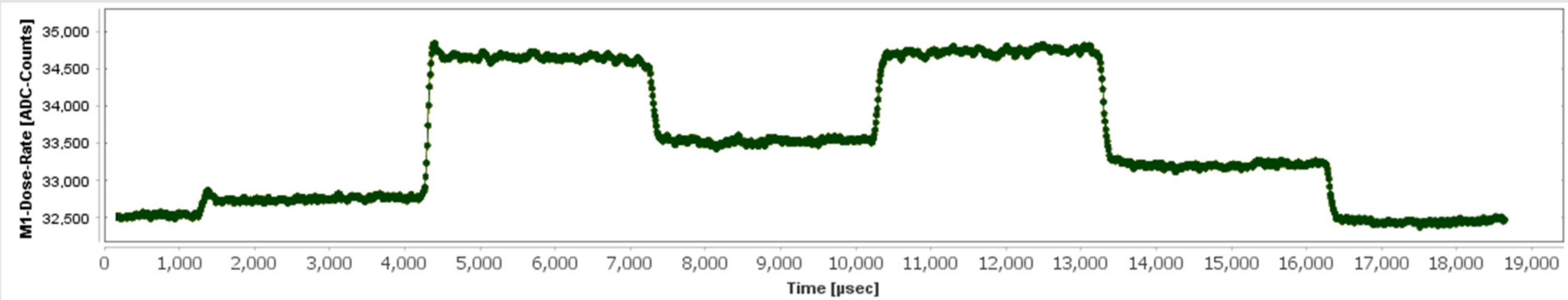
Delivery challenges: Latency

- System latency is 200 us
- Shortest clinical lines expected to be 300 us
- No reaction possible to power supply internal overshoots



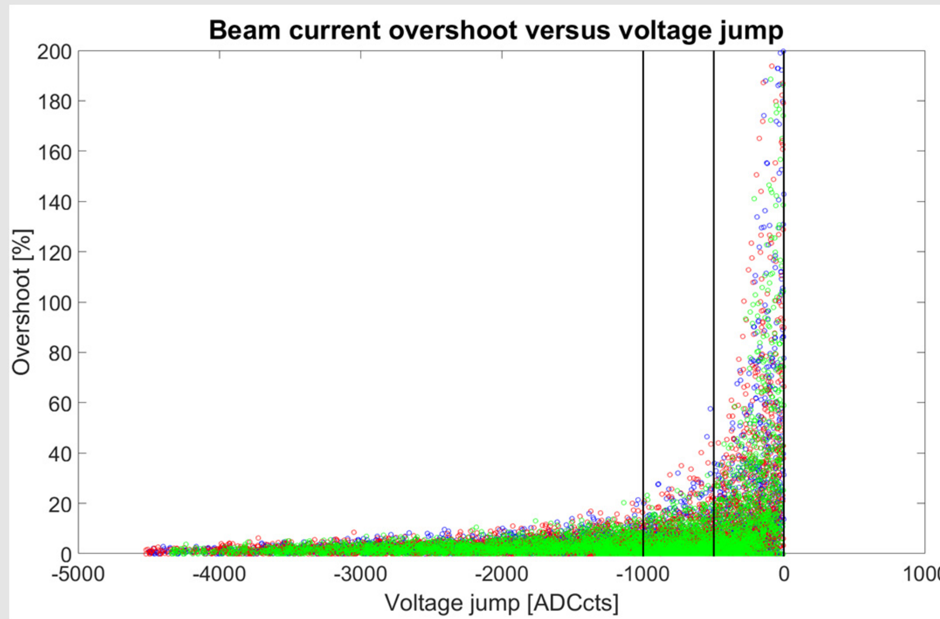
Facility modelling

1. Generate random steps at the VD power supply and record the beam current

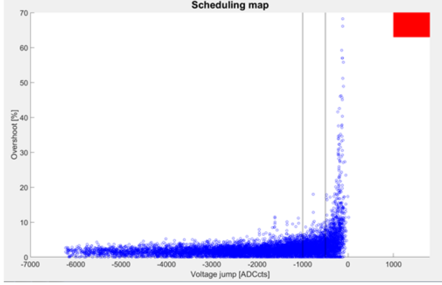


2. Identify transitions and classify:
   – Voltage jump
   – Overshoot = $\frac{Maximum\ intensity}{average\ stable\ value}$

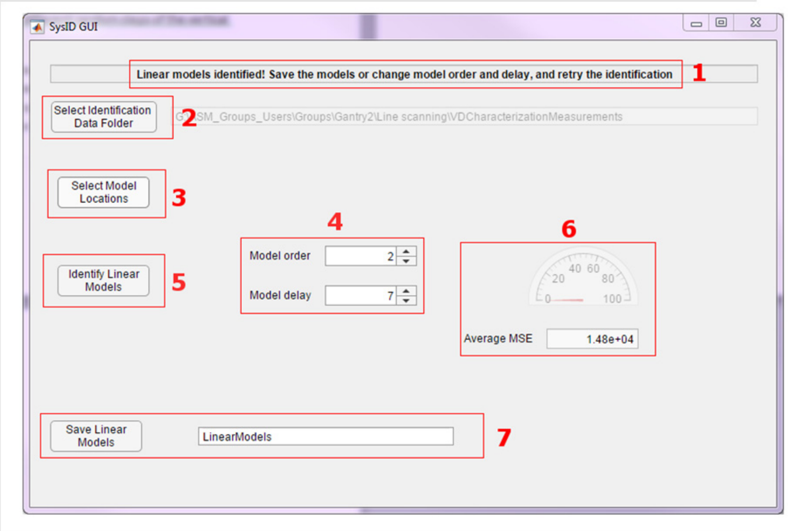
3. Group jumps into low, medium and high overshoot potential



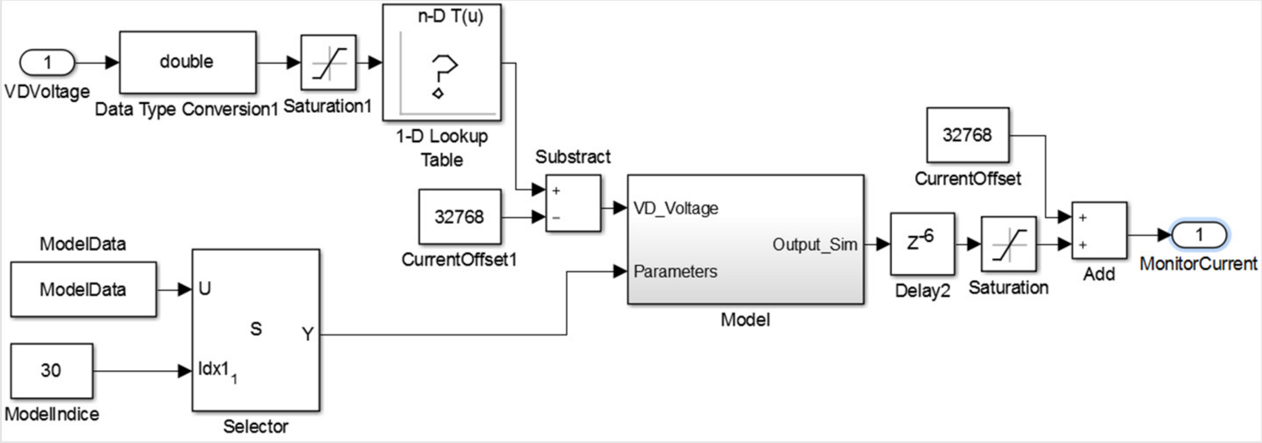
Facility modelling (II)



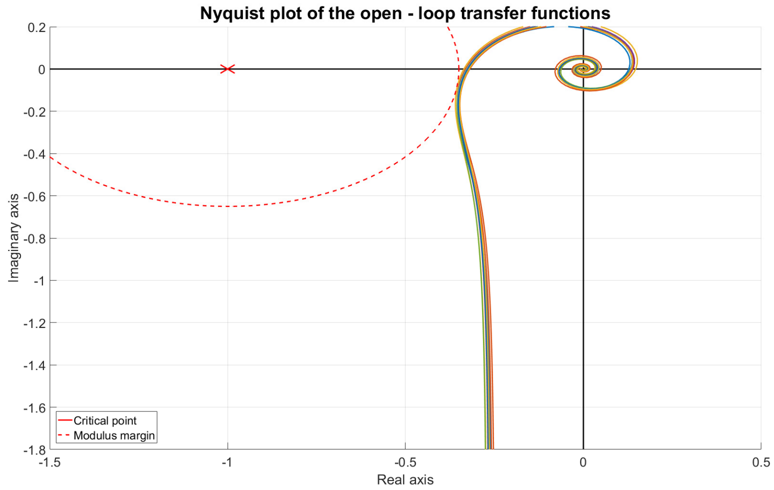
- Matlab GUI to load measurement files
  - As many files from different days as possible
  - Select model parameters
  - Choose representative points in the plot, one per averaged lineal model desired



- Simulink model created is a collection of linear models, automatically chosen at simulation



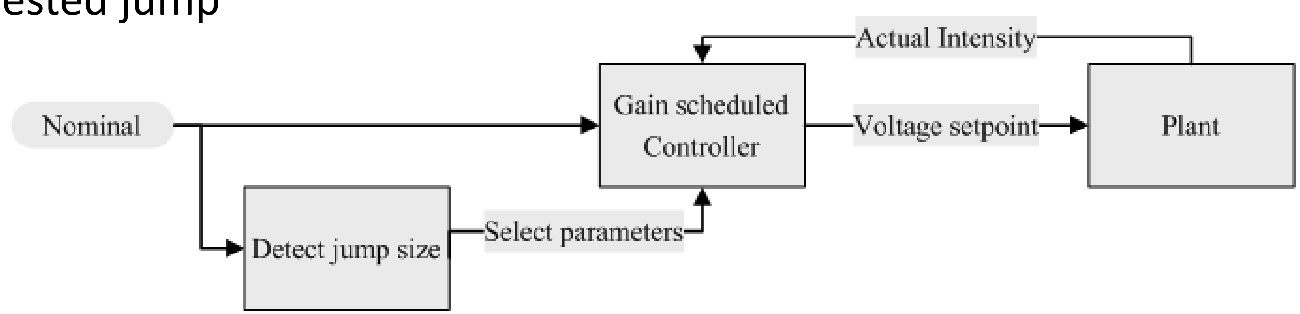
Controller design



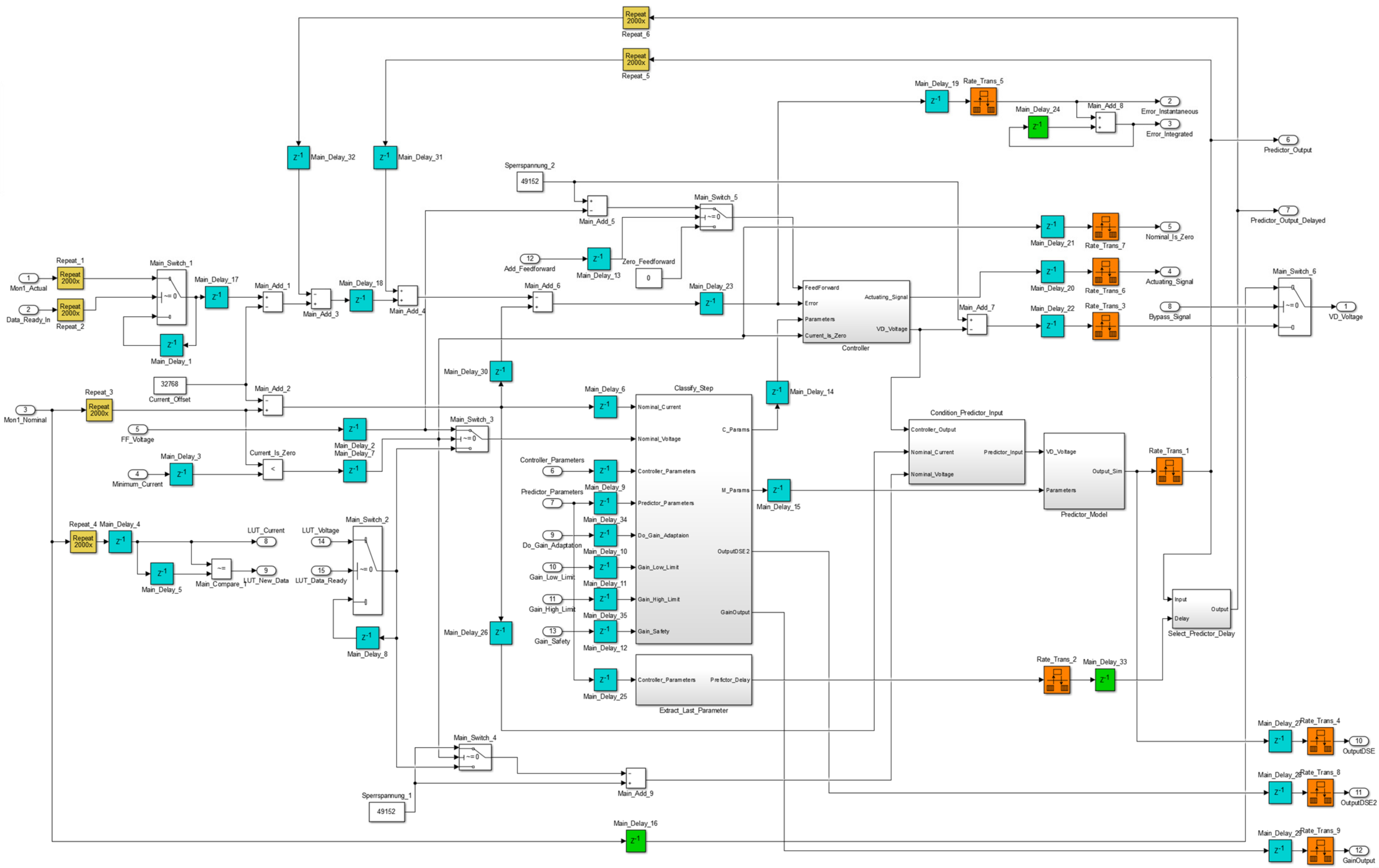
- Use frequency-domain robust control (FDRC)
  - Define an ideal desired close loop to fulfill design requirements
  - Specify desired robustness (modulus margin)
  - Iterate using the facility model until met

- Gain scheduled controller:
  - Small jumps: high potential overshoot→ slow, conservative controller
  - Medium jumps: medium potential overshoot→ intermediate controller
  - Large jumps: Low potential overshoot→ fast, aggressive controller

- The controller will select online the optimal parameters depending on the requested jump

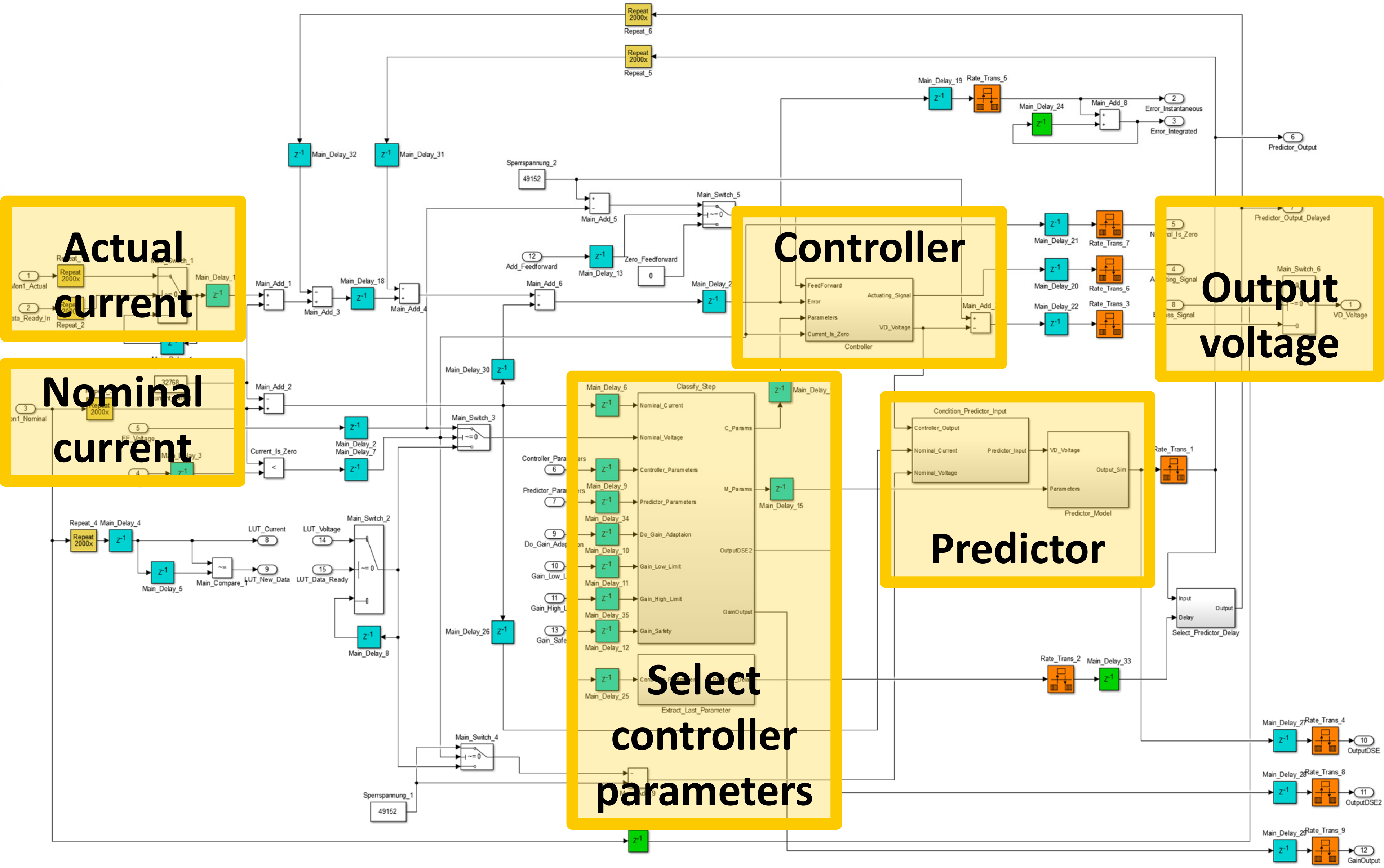


Controller design (II)

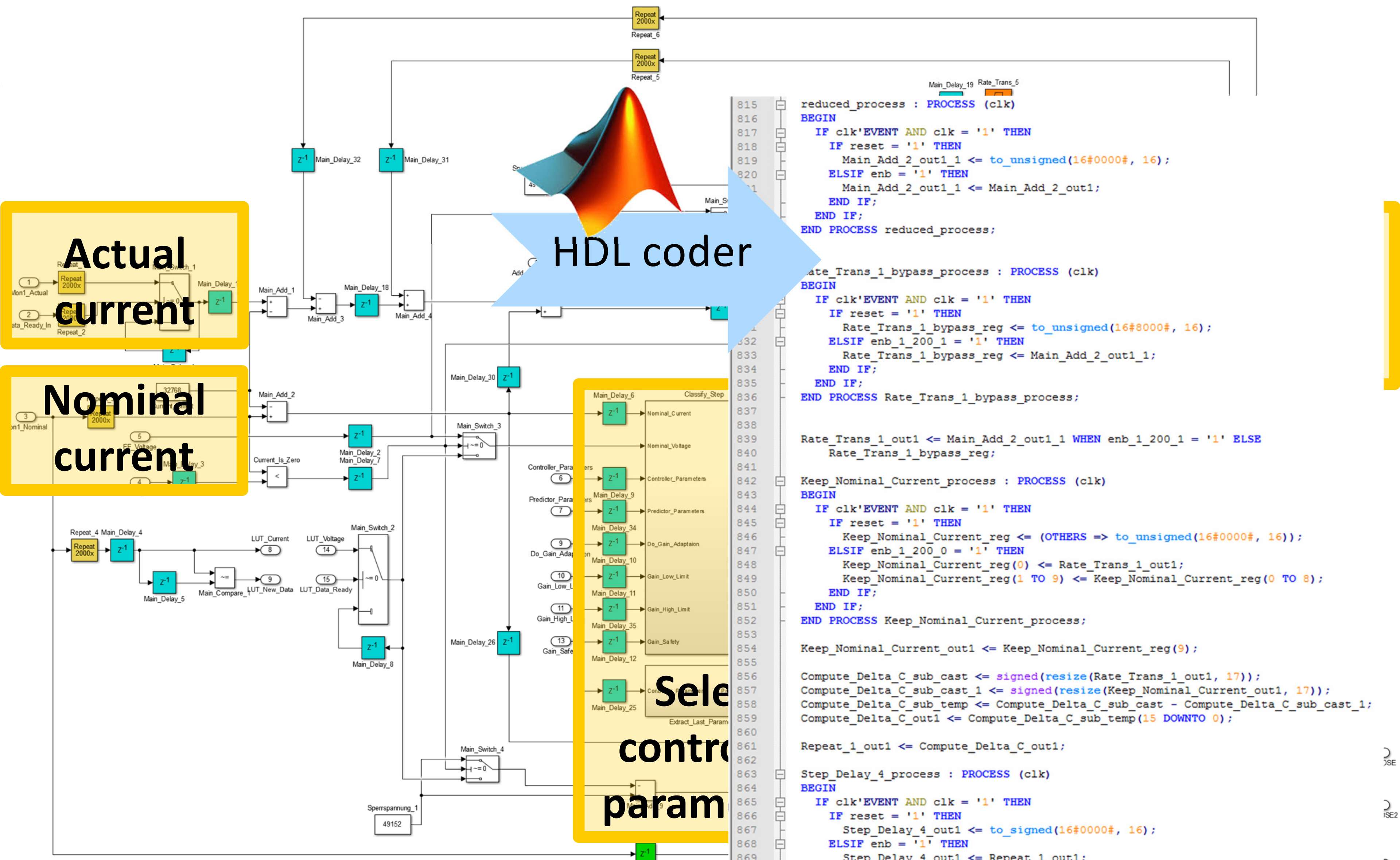


PAUL SCHERRER INSTITUT

Controller design (II)

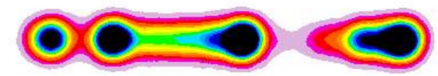


Controller design (II)

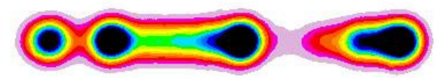



```
reduced_process : PROCESS (clk)
BEGIN
  IF clk'EVENT AND clk = '1' THEN
    IF reset = '1' THEN
      Main_Add_2_out1_1 <= to_unsigned(16#0000#, 16);
    ELSIF enb = '1' THEN
      Main_Add_2_out1_1 <= Main_Add_2_out1;
    END IF;
  END IF;
END PROCESS reduced_process;


ate_Trans_1_bypass_process : PROCESS (clk)
BEGIN
  IF clk'EVENT AND clk = '1' THEN
    IF reset = '1' THEN
      Rate_Trans_1_bypass_reg <= to_unsigned(16#8000#, 16);
    ELSIF enb_1_200_1 = '1' THEN
      Rate_Trans_1_bypass_reg <= Main_Add_2_out1_1;
    END IF;
  END IF;
END PROCESS Rate_Trans_1_bypass_process;


Rate_Trans_1_out1 <= Main_Add_2_out1_1 WHEN enb_1_200_1 = '1' ELSE
  Rate_Trans_1_bypass_reg;

Keep_Nominal_Current_process : PROCESS (clk)
BEGIN
  IF clk'EVENT AND clk = '1' THEN
    IF reset = '1' THEN
      Keep_Nominal_Current_reg <= (OTHERS => to_unsigned(16#0000#, 16));
    ELSIF enb_1_200_0 = '1' THEN
      Keep_Nominal_Current_reg(0) <= Rate_Trans_1_out1;
      Keep_Nominal_Current_reg(1 TO 9) <= Keep_Nominal_Current_reg(0 TO 8);
    END IF;
  END IF;
END PROCESS Keep_Nominal_Current_process;

Keep_Nominal_Current_out1 <= Keep_Nominal_Current_reg(9);

Compute_Delta_C_sub_cast <= signed(resize(Rate_Trans_1_out1, 17));
Compute_Delta_C_sub_cast_1 <= signed(resize(Keep_Nominal_Current_out1, 17));
Compute_Delta_C_sub_temp <= Compute_Delta_C_sub_cast - Compute_Delta_C_sub_cast_1;
Compute_Delta_C_out1 <= Compute_Delta_C_sub_temp(15 DOWNTO 0);

Repeat_1_out1 <= Compute_Delta_C_out1;

Step_Delay_4_process : PROCESS (clk)
BEGIN
  IF clk'EVENT AND clk = '1' THEN
    IF reset = '1' THEN
      Step_Delay_4_out1 <= to_signed(16#0000#, 16);
    ELSIF enb = '1' THEN
      Step_Delay_4_out1 <= Repeat_1_out1;
    END IF;
  END IF;
END PROCESS Step_Delay_4_process;
```

Results: Experiments with beam

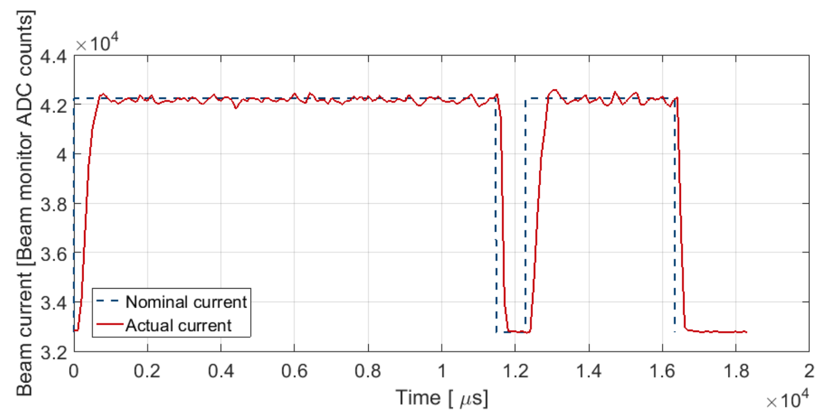


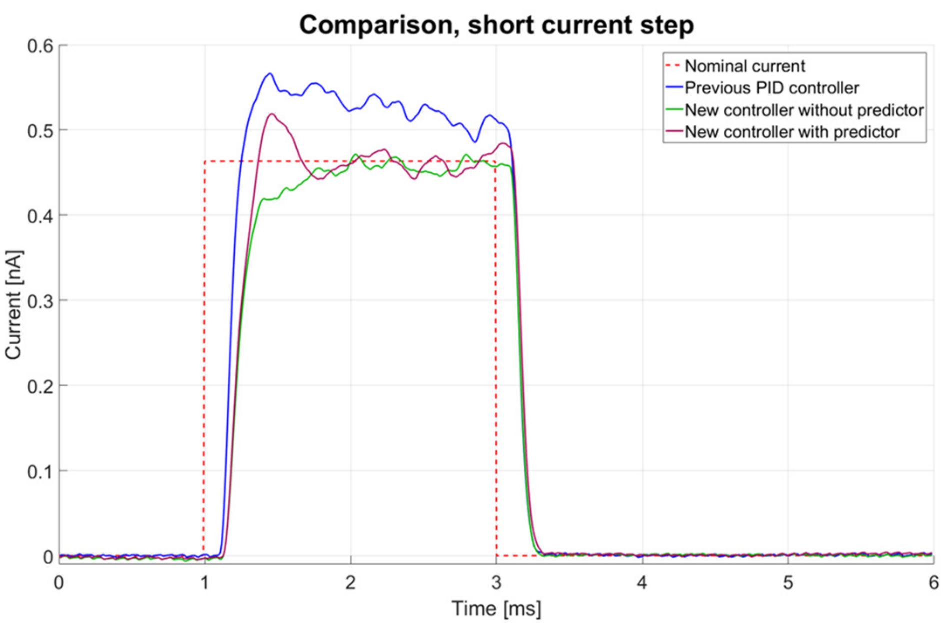
Spot scanning

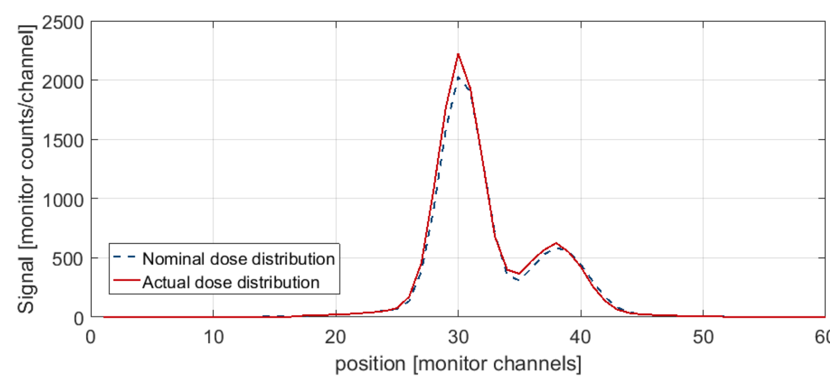


Line scanning

- Dosimetry equivalent to spot scanning (but faster)
- Instantaneous control within specs
- Dosimetry analysis for first fields satisfactory
- Improvement compared to PID
  - Less overshoot
  - Faster setting time







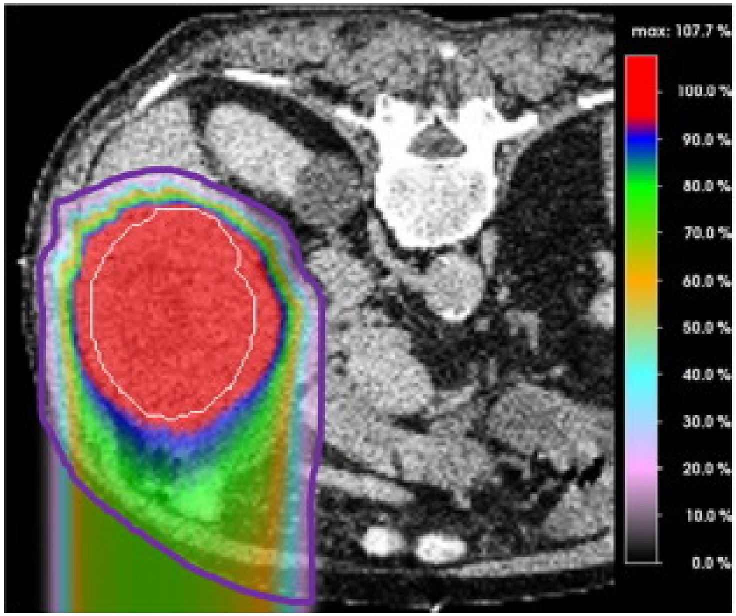
Conclusion

- Modelled the facility using different linear models for different set points
- Used Simulink to generate a gain scheduled controller
- Generated code using HDL coder, and implemented it in the control system FPGA
- Results show better performance than PID controller

Take home message: Simulink + HDL coder is powerful but an understanding of the underlying hardware is required

Outlook

- More testing required before going clinical
  - Find controller parameters that work everyday
  - Or automate characterization of the conditions of the day



Courtesy Cássia O. Ribeiro

Thank you very much

- V. Minnig
- S. Psoroulas
- G. Klimpki
- M. Eichin
- D. Meer
- C. Bula
- D. Weber